Fiscal Year:	FY 2010	**Task Last Updated:**	FY 12/09/2009
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Project Title:	The Role of Intestinal Inflammation in Acute Effects Induced by Exposure to Protons During Solar Particle Events: A Mouse Model Approach		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Radiation Effects Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Project Type:	Ground	**Solicitation / Funding Source:**	2008 NSBRI-RFA-08-03 Postdoctoral Fellowships
Start Date:	11/01/2008	**End Date:**	10/30/2010
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Fornace, Albert (MENTOR/Georgetown University)		
Grant/Contract No.:	NCC 9-58-PF01904		
Performance Goal No.:			
Performance Goal Text:			
	POSTDOCTORAL FELLOWSHIP Sporadic and unpredictably large solar particle events (SPEs) pose serious health risks for future exploratory missions beyond Low-Earth Orbit. During large SPEs, astronauts could be exposed to high doses of high-energy protons and high-Z and high-energy (HZE) particles in a time frame of hours to few days. This acute radiation exposure could cause the prodromal syndrome – a transient period of anorexia, nausea and vomiting that starts within a few hours, and may compromise crew performance. At expected doses, acute effects could be due to direct damage to intestinal mucosa as well as the effect of injury responses that involve intercellular signalling. However, molecular determinants of the prodromal syndrome are mostly still unknown.		

Task Description:

The gastrointestinal (GI) tract is one of the most radiosensitive organs, which is not surprising considering the high turnover rate of mucosal cells. More than 30 yr ago, Potten demonstrated the exquisite sensitivity of intestinal crypt cells. While death by high doses, >10 Gy, in humans is due to GI toxicity, apoptosis of mucosal cells can be seen at 0.5 Gy and less. Several mouse models have been used to investigate early radiation-damage and inflammation-related GI injury responses by gamma-rays. Many studies show an increased expression of pro-inflammatory proteins such as macrophage inflammatory protein-2 (MIP-2) and tumor necrosis factor-alpha (TNF-alpha) in murine or rat intestine after total body irradiation (TBI) of animals. There is evidence that many chemokines are enhanced by inflammatory stimuli. In addition to direct effects, IR-induced cytokines may compromise intestinal function. Our laboratory has seen substantial overlap between the in vivo responses to ionizing radiation (IR) and treatment with lipopolysaccharide (LPS), which triggers an inflammatory response (unpublished data). LPS is also known to affect a variety of important intestinal parameters such as mucosal permeability. If space radiation, such as during large SPEs, triggers a stronger activation of such inflammatory signaling mechanisms, then this may contribute to compromise of GI function.

My principle hypothesis is that inflammatory signaling contributes to the prodromal syndrome and can affect the gut either by local signaling events and/or by systemic cytokine signaling. I am using a genetic approach in a defined mouse model system to study inflammatory signaling after space radiation with emphasis on protons, and the effects of such signaling on cellular and molecular parameters in the intestine. By blocking p38 signaling with a dominant-negative p38 mutant, generated in the laboratory of my mentor Dr. Albert J. Fornace Jr., my intent is to assess the potential advantages of anti-inflammatory therapies to treat the prodromal syndrome.

Specific Aim 1: To measure acute changes and inflammation-associated responses induced in the small intestine of C57BL/6J (BL6) wild type (wt) mice by total body irradiation (TBI) exposure to protons (also in simulated SPEs),HZE ions, or gamma-rays.

Approach Aim 1: The initial effort is to compare the biological effectiveness of high-energy protons and HZE to gamma-rays in wt mice for a variety of cellular and molecular endpoints related to inflammation-related signaling.

Main findings Aim 1: Several molecular parameters appear to be differently regulated in gamma-irradiated animals when compared to equitoxic doses of 1 GeV/n protons and 1 GeV/n 56Fe ions exposed mice. Pro-inflammatory effects, at both systemic and local level, appeared to be greater following exposure to HZE particles and protons.

Specific Aim 2: To determine the involvement of p38 signaling in the acute intestinal and systemic responses to radiation, particularly in simulated large SPEs, and to discern signal transduction pathways involved in these responses.

Approach Aim 2: We will employ a genetic approach with our dominant-negative p38 mutant mouse model. Mice are being bred and litters genotyped at Georgetown University. Our p38 mutant can be used as a model for countermeasures with anti-inflammatory agents to block adverse effects of radiation signaling. Based on results of Aim 1, we will choose two doses for each radiation type (gamma-rays, 1 GeV/n protons and simulated SPEs, 1 GeV/n 56Fe ions) and carry out the same time points of Aim 1 in our p38a+/DN mouse model. Based on our preliminary findings, we expect the p38a+/DN mouse model to show a reduced inflammatory response, which will be reflected at the gene expression and protein level.

Main Findings Aim 2: Although the effort for the objective proposed in Aim 2 was planned to commence at the beginning of year 2, we have already conducted a first set of exposures to gamma-rays and analyses are ongoing for endpoint related to inflammation, proliferation and signaling at both local and systemic level. Our findings support the hypothesis of the original proposal. Exposure to HZE and high-energy protons appears to induce a more pronounced response at several levels, both locally in the intestine and systemically in wt mice.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Radiation-induced prodromal syndrome is a distressing and clinically important adverse outcome in patients undergoing radiotherapy. Moreover, also in the case of individuals involved in radiological accidents the same symptoms may appear within few hours of the exposure. Although patients receiving TBI are at greatest risk, prodromal syndrome is also seen following partial body irradiation and abdominal irradiation. Missing effective prophylaxis and/or treatment, affected patients experience various complications, including dehydration, electrolyte unbalance and malnutrition, which all may delay disease management and hamper quality of life in these subjects. The incidence, severity and onset of symptoms, such as nausea and vomiting, depend on site of irradiation, dose, and size of targeted field. Although the molecular and cellular mechanisms underlying the prodromal syndrome are still unclear, both gastrointestinal effects and central effects are likely to be involved. Our research is aimed to contribute not only in the prevention and/or management of acute space radiation effects, but also in the treatment of prodromal sydrome in patients undergoing radiotherapy or in individuals involved in radiological accidents. The tryptophan derivative 5-hydroxytryptamine (5-HT, or serotonin) is an important signaling molecule in the brain and periphery. The GI tract is where 95% of the total body serotonin is produced and stored. Upon radiation exposure, enterochromaffin-like cells in the intestine release serotonin and this event triggers nausea and vomiting via peripheral and central mechanisms involving 5-HT3 receptors, visceral afferent fibers and chemoreceptor trigger zone (CTZ). In vitro evidence suggests that p38 MAPK is involved in serotonin transport through activation of serotonin transporters (SERTs). Studies employing multiple p38 MAPK activators (ansyomicin, H2O2 and UV radiation) showed an elevation of 5-HT transport in parallel with p38 MAPK phosphorylation, as well as suppression of anisomycin stimulation by p38 MAPK siRNA treatments.
We hypothesize that p38 MAPK is a critical component in up-regulation of serotonin transport in the intestine and that may contribute to onset of prodromal syndrome, Therefore, we are interested in investigating the joint effect of radiation exposure and p38 activity on serotonin release and transport both locally in the GI tract and at a systemic level. The identification of molecular determinants of the prodromal syndrome will importantly contribute in development of pharmacological treatment to be employed not only for space risk countermeasures, but also on Earth in clinically and accidentally exposed individuals.

Task Progress:	Overall, during the course of the first year of the project substantial progress was made for both Aim 1 and Aim 2. Our principle hypothesis is that p38 dependent inflammatory signaling contributes to the prodromal syndrome and can affect intestinal status either by local signaling events and/or by systemic cytokine signaling. The main focus of this research is early alteration of oxidative stress, cytokine signaling and transcriptional profiling locally in the small and large intestine following exposure to gamma-rays vs. space radiation. A limited analysis of systemic response is also being performed. - In wt mice, dose- and radiation-type dependent increase in ROS level was observed that suggests pro-oxidant inflammatory-like response is more accentuated for space radiation than for gamma-rays. - We are observing a central role of p38 in inflammation-associated signaling after IR exposure for both gamma-rays and space radiation: inflammation and cytokine associated genes show attenuated responsiveness compared to wt by expression profiling. - We are investigating the different molecular responses separately for the different intestinal segments (i.e. duodenum, jejunum, ileum, colon) in order to account for specific responses within the intestinal tract. Furthermore, specific intestinal epithelial response is also being investigated on samples obtained by isolation of the intestinal epithelial cells. - We are exploring local and systemic regulation of selected neurotransmitter synthesis and trafficking pathways, which may play an important role in acute response to moderate dose of space radiation. - Staining for proliferation markers, such as PCNA, indicates a role of p38 MAP kinase in cellular proliferation as early as 24 hr after radiation exposure. The research for the second year will aim to add more molecular insight of the differential effects induced by proton vs. gamma-ray exposure at both transcriptional and signaling level in the small intestine. Furthermore, we will continue investigating systemic response in body fluids such as serum and urine. As described in Aim 2 of the original proposal, based on results of Aim 1, we will choose two doses for protons, HZE and gamma-rays and carry out the same time points of Aim 1 in our p38a+/DN mouse model. Markers found in samples from mice exposed to protons and HZE or gamma-rays, will be compared to the lipopolysaccharide (LPS) response to assess similarities. We have observed significant overlap between ionizing radiation IR and LPS treatment at the transcriptomics and metabolomics levels following exposure to gamma-ray (unpublished data). Selected serum samples will be similarly analyzed to assess cytokine levels and will be compared with results from our LPS treated cohort. This is an important step in our project: detection of inflammatory cytokines in the small intestine vs. serum will allow us to assess local vs. systemic induced effects.
Bibliography Type:	Description: (Last Updated: 11/13/2018)
Awards	Trani D. "NASA Student Travel Award to attend the Heavy Ions in Therapy and Space Symposium 2009 for: Increase in intestinal tumors and sustained oxidative stress with Space Radiation. Proceedings of the Heavy Ions in Therapy and Space Symposium 2009, Cologne, Germany, July 6-10, 2009." Jul-2009
Awards	Trani D. "Radiation Research Society (RRS) Scholar in Training (SIT) Travel Award to attend the 55th Annual Meeting of the RRS, Savannah, GA, USA, October 3-7, 2009 for: The role of intestinal inflammation in acute effects induced by exposure to protons during solar particle events (SPEs): a mouse model approach." Oct-2009